Pankaj Rana

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Nationality Indian (Student Visa)
D.O.B 14 September, 1996

LinkedIn

LeetCode

Professional Experience

May 2025 – Feb. 2026

Master's Thesis Student at Zentrum für Sonnenenergie- und Wasserstoff-Forschung Baden-Württemberg (ZSW)

Project Real-time Bat detection on wind turbines | 3D Object detection with AI

- Developed C++ modules for real-time LiDAR data acquisition, preprocessing, and 3D point cloud filtering within ROS2.
- Implemented low-latency background subtraction and clustering-based object detection optimized for outdoor deployment.
- Designed and integrated sensor calibration routines for synchronized LiDAR and NIR operation in field environments.
- Contributed to real-time perception pipelines with GPU-accelerated processing and performance-aware system design.

Aug. 2024 – Feb. 2026

Research Assistant at the Chair of Dynamics and Control (SRS), University of Duisburg-Essen

Computer Science, DSA, and Programming

- Developed programming exercises in data structures and algorithms (DSA), applying object-oriented programming principles.
- Designed test cases to validate correctness and performance of solutions.

Nov. 2024 – April 2025

Software Development Intern at Porsche Engineering Services

Project ADAS Training Simulator | Software Development and Autonomous Driving

- Developed a Windows application in C++ to control the steering wheel, seat actuator, and other hardware components of the ADAS training simulator.
- Implemented real-time control loops using inputs from the CARLA simulation environment to drive hardware behavior.
- Worked in an agile development environment following Scrum and Kanban methodologies to ensure iterative delivery and system reliability.

June 2024 - Sept. 2024

Research Assistant at DST - Development Center for Ship Technology and Transport Systems e.V.

Project Cobotank | Robotics

- Developed C++ modules for coordinate transformations in the robot base frame.
- Conducted system debugging and error diagnosis to improve software reliability.
- Used Git for version control to collaborate and track code changes.

Education

April 2022 - Present

Master's Degree (M.Sc.) University of Duisburg-Essen, Germany

Mechanical Engineering (Specialization in Mechatronics)

- Embedded & Control Systems
- Sensors & Actuators
- Programming in C/C++
- Current grade: 2.1

Projects

Oct. 2023 - Jan 2024 **Board Plotter – ESP32 Microcontroller (Embedded C++)**

University of Duisburg-Essen

- Programmed the ESP32 microcontroller in C++ to control a cable-driven plotter.
- Development and implementation of precise motion logic and algorithms.
- Configuration, calibration and ensuring the accuracy of the motors.
- Integration of safety mechanisms and error handling.
- Carrying out tests and optimizations to improve performance.

Aug. 2016 - July 2020 **Bachelor of Engineering (B.E.) Panjab University, India**

Mechanical Engineering

- Control Systems & Programming Fundamentals
- Final grade: 2.25

Skills

Language Skills **German** (B2), **English** (C2)

Programming & Embedded Systems

- C/C++, Python
- Microcontroller programming (ESP32, embedded C++)
- Real-time control software
- Firmware development & debugging

Tools & Platforms

- Linux
- Bash/Shell Scripting
- GitHub/ GitLab
- Jira, Confluence
- CMake, Docker (Basics)
- MATLAB, Simulink

Development Practices

- Object-Oriented Programming (OOP)
- Unit Testing & Debugging
- Performance optimization for embedded systems
- V-Modell
- Agile (Scrum, Kanban)